

IT Project Proposal

Modelling Natural Disasters In Selangor

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Abstract

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This document is a proposal detailing our group's plans to build our IT project on modelling natural disasters in Selangor. The justification for such a project is the recurring threat that natural disasters pose to our country and its citizen's wellbeing, of particular concern being seasonal floods and flash floods. It is our belief that the data analysis and machine learning skills we have studied over the course of our program will be able to help alleviate this threat to a certain extent.

More concretely, our proposed project's objective is to create a map of Selangor that makes use of machine learning and data analytics in order to predict the likelihood and severity of natural disasters happening within the area. The project's primary focus is on flooding, with additional analysis of disasters such as landslides and earthquakes being treated as supplementary information. The completed model and map will then be hosted on a website or data notebook to facilitate the sharing of the information contained within it.

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1.0 Introduction

Towards the end of every year, Malaysian citizens routinely face harsh floods that greatly affect their livelihoods and general well-being. In 2021, the state of Selangor recorded an unprecedented number of monetary losses caused from these annual floods, totalling to around RM 3.1 billion (Malaysia, Department of Statistics 2021). As a highly industrialized area that consistently boasts the highest total Gross Domestic Product (GDP) out of all states since 2017 (Malaysia, Department of Statistics 2017), Selangor is of vital importance to the Malaysian economy and as such should be allowed to prosper without the threat of natural disasters.

While in recent years there have been many efforts by the Malaysian government to alleviate this issue, most notable among them being the National Flood Forecasting and Warning Program slated to be completed by 2025 (Malaysia, Department of Irrigation and Drainage 2022). This system should be able to model and forecast floods that occur within the entirety of Malaysia through a comprehensive system of sensors and satellites that gather relevant data into a centralized database. However, we believe that we can make use of both openly available datasets as well as data gathered through web scraping in order to achieve a similar, if albeit more limited, system.

Thus, the objective of this project is to deploy a model that can forecast the probability of a flood within an area and show its findings through an easily interpretable visualization such as a geographic map so that more people can become aware of the severity of this issue and that the relevant authorities can streamline their efforts based on the predictions generated. While our primary interest is in modelling flooding, related disasters such as landslides and earthquakes will also be included as supporting data as long as it is possible to do so.

This document shall detail our proposed project through six different sections, with each focused on the problem statement, objectives, scope of work, methodology, proposed outcome and timeline, and risk management respectively. A brief description of each section can be found below:

- **Problem Statement:** A detailed analysis on the background and other related information surrounding the problem, as well as a formal description of the problem itself.
- **Objective:** A brief on the purpose of this project and the potential benefits of its creation.
- **Scope of Work:** A look at the boundaries and restraints placed on the project, both to clearly define its minimum requirements as well as to avoid taking on more work than is advisable.
- **Methodology:** A step-by-step look at the project from the perspective of modern data science methodologies and common practices.
- **Proposed Outcome and Timeline:** A brief on the project's final outcomes and deliverables, as well as a description of its timeline with a supporting Gantt Chart.
- **Risk Management:** A thorough description of possible risk factors of the project, and the strategies put in place to mitigate said risks.

2.0 Problem Statement

The first recorded flood in Malaysia was in the year 1969. In the latter part of December of the previous year, Malaysia was hit by one of the worst floods in the history of the country. Because of the flooding, 70,000 residents in the state of Selangor were forced to abandon their homes, and 25 people ultimately

lost their lives as a result of the disaster. A number of people were left trapped on streets as well as on the roofs of their houses (SoyaCincau, n.d.). It was declared by Izham Hashim, chairman of the state committee on infrastructure and agriculture, that a new flood warning system would be operational by the end of the month of March. It is said that it can provide accurate predictions of floods seven days in advance. This will make it possible to plan things out in more detail and to activate search and rescue teams (BUNYAN, 2022). On September 15, 2022, Director-general Muhammad Helmi Abdullah said that the flood could not speculate on when it would come, whether it will be towards the end of November or the middle of November, or whatever month it will be. (Shah, 2022.)

2.1 Initial Literature Review

2.1.1 Machine Learning

- **Machine Learning**

Machine learning (ML) is a subset of artificial intelligence (AI) that enables software programmes to improve their accuracy at predicting events even if they were not specifically intended to do so. This sort of AI is also known as "deep learning." In order to make accurate projections of future output values, machine learning algorithms need past data as an input (Burns, 2021).

- **Supervised Learning**

A technique to the development of artificial intelligence (AI) known as supervised learning involves the training of a computer algorithm using input data that has been labelled with the characteristics of a desired output. In this method, the algorithm is given data that has not been labelled, and it is programmed to recognise patterns or similarities without any human assistance (SearchEnterpriseAI, n.d.).

- **Linear Regression**

A well-known kind of supervised machine learning model is known as linear regression. In linear regression, the model searches for the linear line that provides the greatest fit between the independent variable and the dependent variable. There are two different types: the basic type and the many types. When there is only one independent variable, a simple linear regression is performed since the model must determine the linear relationship between the independent variable and the dependent variable. Whereas, in Multiple Linear Regression, there are more than one independent variables for the model to include while attempting to determine the connection between the two things (deepanshi, 2021).

2.1.2 Classification Algorithm

A statistical analysis technique called logistic regression uses previous data set observations as well as the information provided above as stated is to predict a binary result, such as yes or no. By examining the connection between one or more already present independent variables, it makes a prediction about a dependent data variable. A logistic regression might be used, for instance, to forecast whether a political candidate will succeed or fail in a campaign or if a professor or lecturer would be admitted to the institution of his or her choice. These simple choices between two options allow for binary results. It

considers a variety of factors, including the number of seats the candidate has won throughout all states in the nation or the professor's or lecturer's highest academic degree. It will use previously collected data to rank incoming cases according to their likelihood of falling into one of two outcome groups. The algorithms improve at predicting classes within data sets as more relevant data is included (Lawton, 2022).

2.1.3 Programming Language Considerations

Python

Python is the best solution for machine learning and AI-based projects because it is simple and consistent, it provides access to great libraries and frameworks for AI and machine learning (ML), it is flexible, it is independent of platform, and it has a larger community. These benefits make Python the best solution. Python's simplicity allows developers to write reliable systems (steelkiwi.com, n.d.).

- It is easy to learn.
- It is easy to understand Python code, making it easy to build models for machine learning.
- Allows to quickly create prototypes for testing your product for machine learning purposes.

There is a disadvantage, and as a result, it is not appropriate for the development of mobile applications (Eastwood, 2020).

C++

"C with Classes" is a frequent nickname for the programming language C++, which is an extension of the C language. C++ is an expansion of the C computer language and also a general-purpose programming language (GeeksforGeeks, 2020). The language has seen significant evolution over the years, and modern C++ currently includes concepts such as object-oriented programming, generic programming, and functional programming in addition to tools for low-level memory management. There is code written in C++ that underpins many of the technologies that are implemented in Python in the form of libraries (Codete Blog, 2021).

- A more efficient and space-saving runtime code.
- Once compiled, C++ runs more quickly.
- There are no complications with the installation.

Since C++ does not include a garbage collector, unneeded data is not filtered away, which may result in a significant waste of storage space. Because of its high memory consumption and other security flaws, programmes built in C++ are not as reliable as those produced in other programming languages (KO2 Recruitment, 2021).

R

When it comes to statistical modelling and analysis, the computer language R is the one that is used the most often. The inclusion of statistical information into one's line of work necessitates the usage of this programming language, which finds widespread use among statisticians and academics. Users are

given the ability to handle and view data via the use of the R programming language, which is also highly suitable for the process of producing data visualisations and visuals. Facebook, Twitter, and Microsoft are just few of the companies that are possible examples of businesses that may employ R Programming as the programming language for their system (Bhalla, n.d.)

- Because it is an open-source language, anybody may create new applications with it or even fix existing problems.
- The charting and graphing processes may be made much simpler by using R
- R are capable of converting unstructured, jumbled data into a more organised format.

When compared to other programming languages such as Python, R requires a greater amount of memory due to the fact that its data are kept in physical memory. The memory stores R's whole data set in a single location for easy access at any time. Which is not the best choice when we are dealing with large amounts of data. Additionally, compared to other programming languages, it moves at a glacial pace (www.javatpoint.com, n.d.).

Therefore, after much deliberation, we came to the conclusion that Python would be the best choice because it is simpler to learn, runs more quickly, offers greater memory flexibility, and is capable of processing Big Data.

2.1.4 Python Libraries

Python libraries are a collection of useful pre-combined programming functions that eliminate the need to manually construct long lines of code. These libraries may be found on a Python computer. When it becomes necessary, they may be brought in from storage. As of present moment, there are about 137,000 different Python libraries that may be used (Team, 2022).

- **Spark SQL**

A programming module for the processing of structured data is known as Spark SQL. It is also capable of performing the functions of a distributed SQL query engine (Spark SQL, 2022). Despite having only one interface, it enables the loading of data from a variety of different sources.

- **Spark ML**

Spark ML is the major machine learning application programming interface (API) that is used for Spark. It is a scalable Python library that contains classification and regression techniques. This library is capable of integrating without any problems with other Spark components, such as the Spark SQL component that was discussed before. In addition to enabling the development of pipelines that are built on top of data frames, Spark ML enables the pre-processing of big datasets, the training of models, and the formulation of predictions based on those datasets (ProjectPro, n.d.).

2.1.5 Natural Disasters and Natural Disaster Prediction Systems

Natural disaster prediction systems have also been of extreme interest to any country at risk of suffering from them. While, Malaysia is quite fortunate to be outside the reach of several highly dangerous natural disasters such as volcanoes, cyclones, tornadoes, and most earthquakes, Malaysia is nonetheless still vulnerable to natural disasters. Notable examples of such vulnerabilities are floods, of both the monsoonal and flash varieties, some earthquakes, as well as tsunamis caused by those earthquakes.

The primary focus of this project shall be floods, however other disasters such as landslides and earthquakes will be considered whenever possible. It can be estimated that Malaysia as a whole suffers billions of Ringgits in damages due to natural disasters, as flooding alone has caused RM3.1 billion in terms of damages in the year 2021 (Malaysia, Department of Statistics 2021). In response to this, the Malaysian government has made several initiatives to help predict and reduce the damage of natural disasters. These responses have been guided by an understanding of the specific natural disaster and related geographical phenomena. For example, data that is relevant for predicting floods are the amount of rainfall in a given area, the rate of change in water level, the type of precipitation that is causing the increase in water level, as well as many other factors (Usgs.gov, 2018).

Natural disaster prediction systems refer to interconnected systems that consist of sensors, models, computers, and other complex components that are designed to help warn the relevant authorities of impending disasters. The following are several relevant examples currently being used or are planned to be used in the future in Malaysia:

- Malaysian Tsunami Warning System, already in use (Telegrafia 2016).
- National Flood Forecasting and Warning Program, to be completed by 2025 (Malaysia, Department of Irrigation and Drainage 2022).

Additionally, several studies have been conducted on the topic of predicted floods through data analytics in the past, such as the work of Malek et al. (2020) and Faruq et al. (2020). While both are very detailed and impressive works on the topic, they do not rely on publicly available satellite data in order to construct their models, instead they relied on data supplied by the Malaysian Department of Irrigation and Drainage. From this, we believe a gap in the existing literature is present and that our project can serve to fill said gap by leveraging open-source earth data in our work.

2.1.6 NASA Earth Science Data Systems (ESDS)

The National Aeronautics and Space Administration (NASA) of the United States of America is an extremely well-known organization the field of science and aerospace developments. A development of particular interest to our project would be their Earth Science Data Systems (ESDS) program, that allows the general public to access earth science data gathered by NASA for their own use. Furthermore, the stated goals of this program, which involve advancing open science, advancing open science data systems, and leading research into earth science data, correlate highly with our own project's objectives (NASA 2019).

The ESDS is actually an integrated collection of various data collected by NASA over the course of their operations. Therefore, only selected datasets will be used in our project. Some examples of such

datasets are the Integrated Multi-satellitE Retrievals for Global Precipitation Measurement (IMERG) dataset that contains global precipitation data gathered from NASA's satellites (NASA 2020) or the Sea Level Change datasets that provide datasets about global sea levels and how they change over time (NASA 2015).

While some datasets from the sources listed above may have some minor restrictions on their access, such as requiring an account or request to access, most of the datasets relevant to this project can be accessed without much issue. Due to the scale of the datasets available on NASA's websites, we will need to specify certain subsets of the dataset in order to download and process them in an acceptable time frame.

Additionally, most data stored on the ESDS uses a special file format known as Network Common Data Form (NetCDF), which is a data format maintained by Unidata and designed to support multidimensional scientific data such as complex geographical data (Unidata 2014). Fortunately, multiple Python libraries exist that help load and process NetCDF data, such as the rioxarray library or the netCDF4 library (Hafen 2020).

2.1.7 Website Development and Deployment Methods

A core objective of this project will be to present our findings and models in an easily accessible manner to the general public as well as interested authorities. To do so, we have come up with several deployment strategies for our project, with the most attractive option as of the writing of this proposal being a website deployment. In this vein, we are aware of two methods that we believe to be achievable based on our skill level and circumstances, with those two methods being a GitHub hosted website or a web application created through the Python StreamLit library.

GitHub is commonly used as a method for software developers or any other interested parties to interact and share their code in an easily manageable and modifiable format online. More formally, GitHub is a cloud-based code hosting service owned by GitHub Inc. that uses the highly popular and open-source version control software known as Git. In truth, Git is fully useable as a standalone tool for version control that can be installed on most computer systems in existence (GitHub 2020). While Git's utility in the code writing process cannot be understated, and certainly should be considered for use in the implementation of our project, it is GitHub's ability to allow users to host webpages that is of concern to us.

GitHub pages is a website hosting service that users of GitHub can access as long as they have registered for an account. To actually build the website, we will still need to make use of HTML, CSS, Javascript as well as any additional tools that we may require along the way. Once the website has been created using the languages mentioned above, we need to create a new GitHub repository specifically for our website and upload all of our website's files into it. Finally, the website may be access through the url "<USERNAME>.github.io" (Deb 2022).

On the other hand, StreamLit is an installable Python library that allows users to create web applications for data science purposes within Python itself, without relying on HTML, CSS, or Javascript. Naturally, as the rest of the project makes use of Python and our own familiarity with the language, such a tool would be highly desirable for this project. A key advantage of StreamLit over other deployment methods is that it is designed to support data applications, such as the one being

considered within this project. As noted by Li (2021), StreamLit is a powerful tool that can mimic data visualization tools like Tableau, while being easy to use and having ample resources for us to learn from.

However, while StreamLit is a powerful tool for developing web applications, it is still not able to host web applications on the Internet. Thus, should we choose to use StreamLit for our final deployment, we will likely still need to make use of another hosting service such as GitHub pages to actually host our website.

2.1.8 Data Notebooks

As a backup plan if a website is deemed unfeasible during the actual implementation of our project, we are prepared to use data notebooks such as Jupyter Notebooks, Google Colab notebooks, Databricks notebooks or any other suitable tool in order to showcase our project.

Generally speaking, data notebooks are a commonly used method by which data analysts, data scientist, and other similar personnel share their findings and machine learning models online. The format is while received by the wider analytics community due to its clean appearance, ease of use, built-in environments, and high shareability allowing for the rapid development of analytics tools (Nehme 2022).

Additionally, many services offering such data notebooks also tend to operate on the cloud and may even offer limited access to their cloud computing clusters. Notably, Google Colab and Databricks, both of which we have used prior to this project, offer such functionality. As we lack sufficient computational power of our own to run models through extremely large datasets, the value of such an additional feature cannot be overlooked, and thus will have its role in the development of our project regardless of if we choose to use notebooks as our final deployment or not.

3.0 Objectives

Using our machine learning model written in Python. The primary goal is to enhance the accuracy of the forecast of flooding or even calamities connected to flooding, such as landslides, which are also one of the causes of floods. The 4SS model will be trained and tested by analysing the data acquired from a variety of datasets obtained from social media, historical data on floods that occurred in Selangor, and data scraped from the web. It will do this by keeping a record of the total amount of rainfall that happens based on a real-time basis, the drainage basin of the river, ground temperature, moisture, and other factors (Usqs.gov, 2018). By using our Python-based machine learning model, we will be able to:

1. Reduce the amount of money that was lost; 2. Reduce the amount of money that will need to be spent on flood recovery, repairing houses, and restoring damaged infrastructure.
2. Lower the number of people who pass away; stop further fatalities in Selangor, which has the greatest record of 25
3. An increase in the states safety against natural disasters, because our 4SS system will be available around the clock, it can perform tests with a higher degree of accuracy than people.

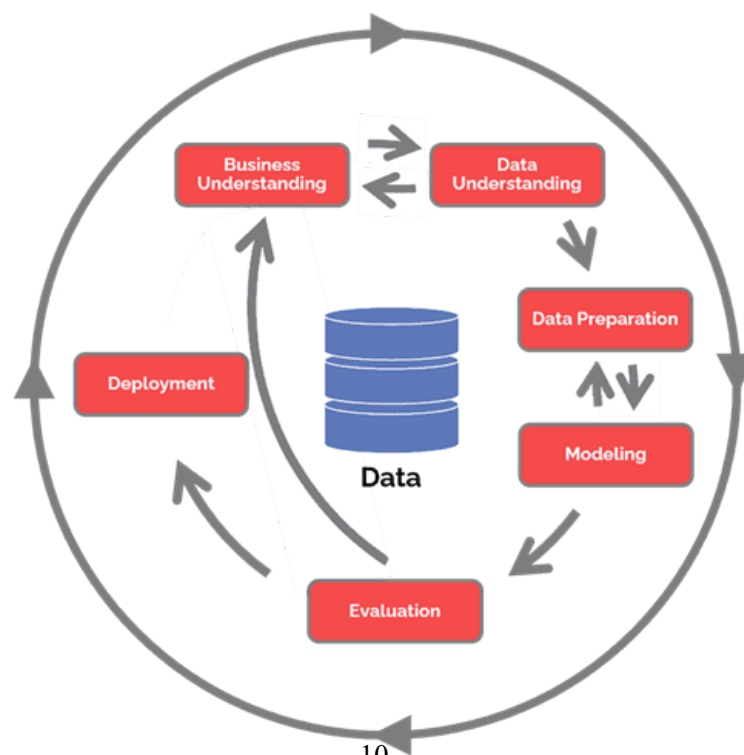
4.0 Scope of Work

4SS itself is the primary deliverable for this project. 4SS is a machine learning model that can classify water levels as either being at their optimal level or being above the level that would cause flooding. 4SS was developed to ensure that the water in Sungai Buloh and Gombak in Selangor is kept at an appropriate level. This is accomplished by incorporating a metre line flood sensor into the system, which detects when the water level rises above the line and sends an alert to warn that a flood is about to occur (Amanda Yeo, n.d.). At this point, the data that has been acquired from these sensors is analysed by our 4SS. Because flood sensor hardware integration and subsequent technologies are outside of the scope of the project and the team's knowledge, the project's limitation is that the focus will only be on the machine learning model after data has been obtained from flood sensors. This is because the flood sensor hardware integration and subsequent technologies are beyond the team's knowledge.

Aside from that, another deliverable is the dataset that will be sent to 4SS for the purpose of training. To guarantee that there is sufficient information for 4SS to train on, this dataset should include an estimated total of more than 400 rows of data. Numerous datasets relating to floods and landslides will be altered and integrated into one major dataset. This main dataset will comprise characteristics based on the domain expertise of the team about water level increases, ground temperature, moisture, and other factors. After that, any errors will be removed from this dataset, and any null values will have averages calculated based on the rows that are still there put in their place. In addition, the dataset will be analysed to evaluate whether it has headers that are accurately labelled with the appropriate information. Since classification is a supervised machine learning strategy, we will be modelling our 4SS using this method. It is necessary for the data that supervised machine learning algorithms utilise to be appropriately labelled for the algorithms to be able to acquire knowledge from examples in the most accurate way possible.

5.0 Methodology

To accomplish our project, we shall closely follow the approach specified in the Cross-Industry Standard Process for Data Mining (CRISP-DM). This is a general process that is frequently used in data science projects regardless of their specific domain. The stages shown in the graphic below shall be elaborated upon and related to our project in the following sub-sections.



Business Understanding

The first phase of the process is understanding the business case of the problem in order to guide future data mining efforts. While this phase is commonly referred to as 'business understanding', it applies to far more than just business, a more accurate term may be 'domain understanding' as the main goal of this phase is to extract the background, objectives, and requirements of the project among other things from the problem of interest's domain.

In this case, this would be building an understanding of flooding and natural disasters in general and from there concretely determine the objectives of our project. A comprehensive literature review of relevant facts about natural disasters and previous studies related to this field will also be performed at this phase. Once a sufficient understanding of the domain and objectives is obtained, a project plan, such as this proposal, detailing the required steps to complete the project will also be prepared.

Data Understanding

The second phase of the process is understanding what data exists or may be gathered in order to solve the problem defined in the previous phase as well as developing an understanding of said data. Typically, this phase involves searching for data sources that may be relevant to the problem and compiling those relevant sources to be further analysed. At this point, Exploratory Data Analysis (EDA) techniques are also used to describe, explore, and verify the collected data.

In this case, relevant datasets such as those from the Malaysian Open Data Portal, Kaggle, or NASA's Global Precipitation Measurement program will be compiled into a comprehensive dataset to be analysed through tools and libraries such as Python's pandas library for data analysis. As mentioned, the goal of this early analysis is to first understand its properties and verify its quality, and then explore it to further understand the dataset in terms of its descriptive statistics, correlations and simple visualizations among many other things.

Data Preparation

The third phase of the process is to prepare the data gathered in the previous phase for modelling by cleaning the data. Famously, this phase is known to take up around 80% of the time allocated to any data science project, and this will likely be the case for our project as well. This is because this phase includes everything from extracting relevant data, cleaning and formatting that data, manipulating and deriving new useful data, and finally integrating it all in a suitable way for modelling.

In this case, we will primarily be using Python for the purposes of this phase. Common data science libraries such as pandas and numpy will no doubt be used extensively, and more specific libraries such as the rioxarray library that will be required for loading and preparing geographic data in the Network Common Data Form (NetCDF) will also be necessary for our project. Depending on the quality of the data gathered, statistical methods to handle missing values and other inconsistencies may also be used here.

Modelling

The fourth phase of the process is to construct a model to solve our problem from the cleaned dataset. The models mentioned in this phase are statistical models designed to perform predictions from a set of input data, with some examples being linear regression and logistic regression. This phase deals with selecting a several suitable models, splitting the data into a training set and a testing set, building the models, and finally assessing which of the selected models performs the best.

In this case, we will primarily be using Python for constructing our models, making use of common machine learning libraries such as scikit-learn, TensorFlow, Keras, or any other library depending on the needs at the time. Throughout this phase, we may utilize research and models that have been successful at predicting natural disasters employed by other countries in the past in order to help with our feature selecting and hyperparameter tuning.

Evaluation

The fifth phase of the process is to evaluate the model constructed in the previous phase. While some technical assessments have already been completed in the previous phase, the evaluation of this phase is more so concerned with the model's capabilities in regard to the business case identified in the first phase. This phase may include evaluating the results obtained through the model using domain knowledge, as well as reviewing the process used and determining the steps need to deploy the model.

In this case, we plan to evaluate the model in terms of its accuracy relative to other similar models in order to judge whether it adequately solves our problem. We will also take care to review whether the process we used to construct our model is valid from the perspective of our chosen domain. After which, we will evaluate and select the best method to deploy the final model by considering the technical requirements of the model such as its required inputs and the form of its outputs.

Deployment

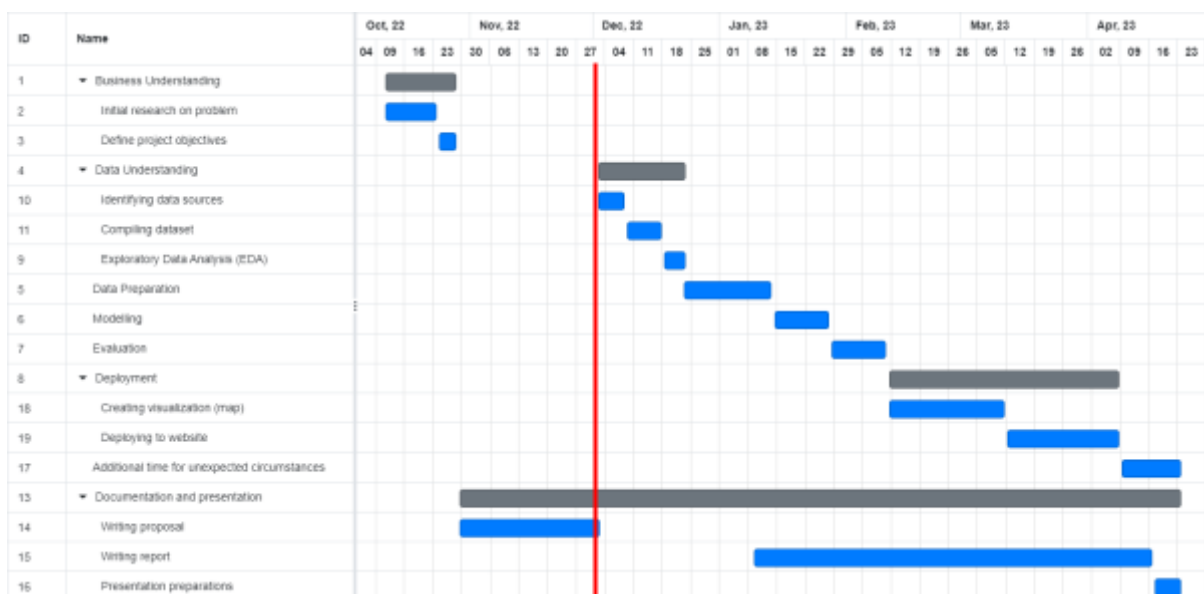
The final phase of the process is deploying the completed model and any accompanying analysis into some easily accessible format. For most cases, this will be a written report with possibly an oral presentation through a slide deck to relevant stakeholders. However, other methods for deployment such as through a website are also viable and may be preferred depending on the specifics of the problem. Regardless of the deployment method selected, monitoring and maintenance plans must be put in place to ensure that the project remains functional throughout the course of its intended use. Finally, a final report of the entire project must also be generated and submitted for review at the conclusion of this phase.

In this case, we plan to deploy our model by graphically representing it as a map, and then launching it on a website supported by either Github or Python's Streamlit library depending on the suitability of each. We have chosen such a method as our goal is to grant as many people as possible access to crucial data about climates and natural disasters in the Selangor region, as well as to help further guide the efforts of authorities. Thus, we believe that the best way to achieve this would be to make our model publicly available and easily accessible to a general audience such as through a website.

6.0 Proposed Outcome and Timeline (with Gantt Chart)

The proposed outcome of this project is a model that can predict the likelihood of a flooding in a given area of Selangor based on precipitation, water level, and other data. The results of this model should be visualized through a map of Selangor, with supporting details where necessary. Finally, the model and visualization should then be deployed on a website or similarly easy to access location.

To ensure that our project proceeds as scheduled, a work schedule in the form of a Gantt Chart has been prepared below.



(Red line separates IT Project Part 1 and IT Project Part 2)

While the Gantt Chart is an extremely useful tool for project management, it fails to capture the iterative cycle that defines the CRISP-DM methodology. Thus, the project schedule above only dictates the time frame that each phase of the process will be primarily occurring in, because it is likely that as the project progresses, certain phases may be revisited to further refine details.

7.0 Risk Management

To prevent avoidable issues cropping up later on due to poor scheduling, we have put in place several contingencies in the event that something unexpected occurs over the course of our project.

As seen in the previous section, we have allocated two weeks of time specifically for unexpected events in our project schedule. It is our hope that this additional “free” time can act as a buffer in the event that our project goes off schedule due to any number of reasons.

Additionally, we plan to organize regular progress checks with our supervisor in order to verify our progress throughout the project duration as well as to consult their expertise on any issues encountered.

In terms of contingencies, our group is prepared to cut out certain parts of the project should they be judged unfeasible for some reason during the execution of the project. Naturally, such cutting will be kept to the absolute minimum and as we see it now, there are two sections of this project that are most at risk of facing technical difficulties.

First, while our initial plan for the project is to model several natural disasters that may occur in Selangor such as floods and landslides among others, we consider flooding the minimum requirement of our project and may cut out other disasters. Second, while we plan to launch our completed project on a simple website to make it easily accessible, should this be unfeasible, we are prepared to use other methods such as a locally run application or a Jupyter Notebook.

The following table details summarizes the strategies put in place for each phase of our project:

Phase	Potential risk	Mitigation strategy
Business understanding	1. Misunderstanding the problem 2. Scheduling issues	For 1) • Consult supervisor on feasibility • Perform thorough research on the domain beforehand For 2) • Allocate additional time for unexpected circumstances • Whenever possible, start tasks earlier than scheduled
Data understanding	1. Failure to locate relevant data sources 2. Difficulty in accessing data sources	For 1) • Perform thorough research for all potential data sources (data.gov.my, Kaggle etc.) • Consider gathering data through web scraping

		For 2) • Consider alternative sources • Consider obtaining access to the data through official channels (email request etc.)
Data Preparation	Technical issues • Messy data • Low quality data • Complications during data processing	• Obtain assistance online or from supervisor • Perform simple feasibility checks in prior stages of the project
Modelling	Low model accuracy	• Repeat training model with different hyperparameters • Attempt different models • Reassess the data gathered
Deployment	Unable to launch website	Utilize alternative deployment methods (Jupyter Notebooks etc.)
Overall project	Data loss from corruption or from computer and/or code malfunctions	Utilize Cloud backups whenever possible and ensuring that both members have their own copies of the work.

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